

Christina Shin

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INTERESTS	Computer Networking, Mobile Computing, AR/VR Systems, Volumetric Video Streaming, Autonomous & Connected Vehicles, 3D Data Processing, 3D Mapping
EDUCATION	University of Southern California, Los Angeles, California ▪ Ph.D. Candidate in Computer Science Jan 2026 (Expected) <i>Dissertation (in progress): Harnessing Vehicular 3D Primitives for Networked Scene Operation and Manipulation</i> Ewha Womans University, Seoul, Republic of Korea ▪ M.S. in Computer Science and Engineering Feb 2019 <i>Thesis: Network Diagnosis and Reconstruction in Vehicular Ad-Hoc Networks</i> ▪ B.S. in Computer Science and Engineering Feb 2017
EXPERIENCE	General Motors, Warren, Michigan Research Intern & Collaborator (Mentors: Fan Bai and Chuan Li) May 2021 – Present ▪ Led the design and implementation of a volumetric video delivery system for vehicles, enabling low-latency and immersive AR passenger experiences. ▪ Developed a 3D traffic reconstruction system using multi-vehicle point cloud registration, generating volumetric videos of dynamic traffic scenes to improve scalability and scene understanding; contributed to a U.S. patent application (US20230140324A1, now abandoned). Networked Systems Lab, University of Southern California Research Assistant (Advisor: Prof. Ramesh Govindan) Aug 2019 – Present ▪ Invented an infrastructure-assisted autonomous driving system that leverages roadside LiDARs and edge compute to extend vehicle perception beyond occlusions. ▪ Devised a 3D building reconstruction system with a LiDAR-equipped UAV, optimizing path planning for data capture and generating near real-time SLAM-based 3D models of buildings. ▪ Contributed to a live volumetric video streaming system, focusing on the 3D video rendering pipeline to ensure real-time visualization of captured scenes. Intelligent Networked Systems Lab, Ewha Womans University Research Assistant (Advisor: Prof. HyungJune Lee) Jan 2017 – May 2019 ▪ Designed a traffic density estimation algorithm using opportunistic V2V packet probing under strict time deadlines. ▪ Proposed a multi-UAV relay algorithm for network recovery, improving resilience of ad-hoc networks in failure scenarios.
PUBLICATION	CONFERENCE ▪ Toward Bandwidth-adaptive Fully-Immersive Volumetric Video Conferencing Rajrup Ghosh, Christina Suyong Shin, Lei Zhang, Muyang Ye, Tao Jin, Harsha V. Madhyastha, Ravi Netravali, Antonio Ortega, Sanjay Rao, Anthony Rowe, Ramesh Govindan <i>International Conference on emerging Networking EXperiments and Technologies (ACM CoNEXT)</i>, 2025 (conditionally accepted).

- **RECAP: 3D Traffic Reconstruction**
Christina Suyong Shin, Weiwu Pang, Chuan Li, Fan Bai, Fawad Ahmad, Jeongyeup Paek, and Ramesh Govindan
International Conference on Mobile Computing and Networking (ACM MobiCom), 2024.
- **Cooperative Infrastructure Perception**
Christina Suyong Shin*, Fawad Ahmad*, Weiwu Pang*, Branden Leong, Pradipta Ghosh, and Ramesh Govindan (* equal contributions)
International Conference on Internet-of-Things Design and Implementation (IEEE/ACM IoTDI), 2024.
- **AeroTraj: Trajectory Planning for Fast and Accurate 3D Reconstruction Using a Drone-based LiDAR**
Fawad Ahmad, **Christina Suyong Shin**, Rajrup Ghosh, John D'Ambrosio, Eugene Chai, Karthikeyan Sundaresan, and Ramesh Govindan
Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (ACM Ubicomp/IMWUT), 2023.
- **Progressive ad-hoc route reconstruction using distributed UAV relays after a large-scale failure**
Christina Suyong Shin, So-Yeon Park, JinYi Yoon, and HyungJune Lee
IEEE Wireless Communications and Networking Conference (IEEE WCNC), 2018.
- **DroneNet+: Adaptive Route Recovery Using Path Stitching of UAVs in Ad-Hoc Networks**
So-Yeon Park, Dahee Jeong, **Christina Suyong Shin**, and HyungJune Lee
IEEE Global Communications Conference (IEEE GLOBECOM), 2017.

JOURNAL

- **Infrastructure-less Vehicle Traffic Density Estimation via Distributed Packet Probing in V2V Network**
Christina Suyong Shin, JiHo Lee, and HyungJune Lee
IEEE Transactions on Vehicular Technology (IEEE TVT), vol. 69, no. 10, Oct 2020.
- **DroneNetX: Network Reconstruction through Connectivity Probing and Relay Deployment by Multiple UAVs in Ad-Hoc Networks**
So-Yeon Park, **Christina Suyong Shin**, Dahee Jeong, and HyungJune Lee
IEEE Transactions on Vehicular Technology (IEEE TVT), vol. 67, no. 11, Nov 2018.

UNDER SUBMISSION

- **Vehicular AR Streaming.** *Submitted to ACM MobiCom 2025.*
- **Vehicular AR Tracking.** *Submitted to ACM MobiCom 2025.*

AWARDS & HONORS

- **Annenberg Fellowship, University of Southern California** 2019
Awarded to outstanding Ph.D. students joining in Fall 2019.
- **Qualcomm Innovation Awards, Qualcomm x Ewha** 2017
Recognized for proposing a lightweight network hole replacement algorithm through UAV-net.
- **Silver Prize in Graduation Capstone Design, Ewha Womans University** 2016
For an outstanding project developing *SimMusic*, a language that encodes music notes and plays them on *Lego Mindstorms NXT*.
- **Dean's List, Ewha Womans University** 2013, 2015, 2016
For attaining a GPA above 3.75/4.3.
- **NSF/SIGMOBILE Travel Grants: ACM MobiCom '25, ACM HotMobile '24.**

TEACHING & MENTORING

TEACHING ASSISTANT

University of Southern California

- Internetworking (CSCI 353) Fall 2025
- Computer Networks (CSCI 551/651) Fall 2023

Ewha Womans University

- Computer Architecture (20493-02) Fall 2018
- Arduino Programming (11208-01) Spring 2018
- C Programming (38407-05) Fall 2017
- Programming Language Theory (20499-01, 20499-02) Spring 2017

MENTORING

- Alejandra Cortes (High school): 3D perception with point clouds Summer 2023
- Alexander Due (Undergrad): Object-oriented point cloud registration Summer 2023
- Zaki Cole (Undergrad): Automated data collection in CARLA simulator Fall 2022

**TECHNICAL
SKILLS**

Languages: C++, Python, MATLAB, C, C#, Java

Systems & Tools: Linux, Git, Docker, ROS, CARLA, AirSim, Unity, PyTorch, TensorFlow